ZHUOHAO WU

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EDUCATION

The Affiliated High School of South China Normal University
Junior high school diploma & High school diploma — Sep 2017 - Jul 2023

Pittsburgh Institute, Sichuan University
Bachelor of Industrial Engineering | **GPA: 3.91/4.0 (Score: 91.25) Rank: 7/73** — Sep 2023 - Jul 2027
Core Curriculum: Probabilistic Methods in Operations Research(99); Reliability Engineering(99); Operations Research(96); Differential Equations(94); Statistical Testing and Regression(94); Productivity, Random Variables, and Distributions(92); Engineering Computing(95); Introduction to Engineering Analysis(94)

RESEARCH INTEREST

- Driver Behaviors and Decision Making in Autonomous Driving
- Human Factors and Human–Computer Interaction
- Large Language Model (LLM) in Healthcare
- Operations Research and Optimization Algorithms

RESEARCH EXPERIENCE

Driver Behavior and Cognitive Psychology of CAVs Based on Driving Simulation — Mar 2025 - Now
Pittsburgh Institute, Sichuan University | Advisor: Prof. Xiaomei Tan & Prof. Yaoyu Fu — *Chengdu, China*

- **Purpose**: Uncovered the decision-making logic and neural mechanisms of drivers' intervention in the CAV context, providing empirical evidence for human-machine collaborative control and safety intervention strategies.
- **Experiment**: Conducted a mixed-design experiment in an autonomous driving simulation study using a 2D virtual city environment, with different unexpected hazards featuring potential conflict points that participants could decide to intervene. 35 participants were recruited to complete the experiment.
- **Data Analysis**: Collected physiological data (fNIRS and EMG) and subjective scale data during the experiment and established statistical models to investigate the underlying neural activities associated with driver trust and intervention tendencies under different autonomous driving contexts.
- One paper was **accepted** by *HFES ASPIRE 2026* (The 70th HFES International Annual Meeting); one paper is prepared to submit to *Accident Analysis and Prevention* (already completed) **(Both First Author)**.

LLMs for Electronic Health Records (EHRs) in Emergency Departments (EDs) — Sep 2025 - Present
Pittsburgh Institute, Sichuan University | Advisors: Prof. Rong Yin & Prof. Peng Deng — *Chengdu, China*

- **Systematic Review**: Under **third-round** peer review at *Journal of Medical Internet Research* (JMIR) **(First Author)**. Systematically summarized and discussed core clinical tasks across emergency care workflows, along with corresponding LLM technical frameworks, evaluation methodologies, and reported performance.
- **Ongoing Work**: Applying LLMs with diverse strategies to validate model capabilities in supporting clinicians on key emergency tasks, including emergency diagnosis and patient disposition prediction, using collected multimodal EHR data and clinical indicators.

Reliability Modeling of Multi-Turn Constraint Adherence in LLM Systems — Mar 2026 - May 2026
Pittsburgh Institute, Sichuan University | Advisor: Prof. Changxi Wang — *Chengdu, China*

- Formulated multi-turn instruction retention as a reliability engineering problem to evaluate whether LLMs can preserve initial constraints as dialogue history and conflicting context accumulate. Built a deterministic rule-based evaluation protocol and evaluated six LLMs on 87 dialogue samples on constraints of Content, Formula and Length under Baseline, Perturbation, Reminder, and Perturbation+Reminder conditions. One paper was accepted by *IEEE RASSE 2026*; **Co-first Author**.

A Multi-Agent-Driven Psychological Healing Game Generation System for CBT Sep 2025 - Jan 2026
College of Computer Science, Sichuan University | Advisor: Prof. Hongru Liang *Chengdu, China*

- By leveraging the multi-modal content generation and multi-modal emotion analysis capabilities of LLMs, establish a multi-agent collaboration game framework for personal clinical practice, including an assessment strategy library, a CBT expert strategy library, and a long-term memory module.

WORK AND ACADEMIC EXPERIENCE

INTR, The Hong Kong University of Science and Technology (Guangzhou) Jul 2026 - Aug 2026
Summer Research Assistant | Adviser: Prof. Dengbo He *Guangzhou, China*

Performance: Conduct and accomplish an eVTOL takover experiment with using eye-tracking, EDA, ECG, and RESP data in a six-degree simulator (on-going project). A manuscript was submitted to *Transportation Research Board Annual Meeting* (Third Author).

The 3rd International Conference on Human Factors Engineering and Human-Intelligence Collaborative Innovation 2026 *Shenzhen, China* | June 2026

Performance: Outstanding Poster Award.

Oriel College, University of Oxford Jul 2025
Summer Institute Course: Symbolic Methods in AI | Lecturer: Prof. Michal Zawidzki *Oxford, UK*

Performance: Course Grade 94 (A+).

SINOTRANS South China Co., Ltd. Jul 2025 - Aug 2025
Innovation & Technology Center Intern *Guangzhou, China*

Performance: Develop a freight price and financial indicator forecasting framework integrating SARIMAX time series models with LLM prompting and fine-tuning techniques to support budgeting and risk management.

AWARDS AND HONORS

Scholarship and Honors

National Scholarship	Dec 2026
Dean's List of Pittsburgh Institute	Oct 2026
Outstanding Student of Sichuan University	Oct 2026
Outstanding Student Leader of Sichuan University	Oct 2025
Sichuan University Comprehensive Second-Class Scholarship (Top 5%)	Oct 2025
Outstanding Student of Sichuan University	Oct 2024
Sichuan University Comprehensive Third-Class Scholarship (Top 8%)	Oct 2024

Academic Competition Awards

- **First Prize (Sichuan Province)**, Contemporary Undergraduate Mathematical Contest in Modeling, 2025.
  Built on kinematic and geometric analyses and constrained optimization theory, applied PSO to develop an optimal UAV smoke-canister deployment strategy under multi-canister time-sequenced coordination and spatial matching in a high-dimensional solution space.
- **Honorable Mention**, Mathematical Contest in Modeling, 2025.
  Built an entropy-weighted, regression-driven multi-objective sustainable tourism model, quantifying economic benefits, resident satisfaction, and environmental pressure linked to visitor travel and transportation emissions, while optimizing under infrastructure and capacity constraints.
- **Second Prize (Sichuan Province)**, The Chinese Mathematics Competitions for College Students, 2024.

SKILLS

Languages: English (IELTS:7; TOEFL: 93), Chinese (Native)
Programming Languages: Python / MATLAB / C++ / C / SQL
Other Skills: • LLM Techniques: Prompt Engineering / Fine-tuning (e.g., LORA) / Design agent systems • Machine Learning Algorithms • Optimization Algorithms • Scientific Writing and Sketching Skills: LaTeX / draw.io / Adobe Illustrator / Microsoft Office Software